From 500 to a General Formula: Exploring a Floor Identity

WANMIN LIU
wanminliu@gmail.com

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1 THE PROBLEM AND A CHANGE OF PERSPECTIVE

The original problem. Prove that there exists a real number w such that

$$\boxed{\sum_{k=0}^{500} \left\lfloor \frac{x+2^k}{2^{k+1}} \right\rfloor = \lfloor x \rfloor - \lfloor wx \rfloor \qquad \text{for every } x \in \mathbb{R}.} \tag{1}$$

Before trying to solve the problem, let us test a special value of x. Taking $x = 2^{501}$, we have

$$\left\lfloor \frac{2^{501}+2^k}{2^{k+1}} \right\rfloor = 2^{500-k}, \qquad 0 \leqslant k \leqslant 500.$$

Therefore,

$$\sum_{k=0}^{500} \left\lfloor \frac{2^{501}+2^k}{2^{k+1}} \right\rfloor = \sum_{k=0}^{500} 2^{500-k} = 2^{501} - 1.$$

If the identity holds for every real x, we must have

$$2^{501} - 1 = 2^{501} - \left\lfloor 2^{501} w \right\rfloor,$$

and hence

$$\left\lfloor 2^{501} w \right\rfloor = 1.$$

By the definition of the floor function,

$$1 \leqslant 2^{501} w < 2.$$

Dividing by the positive number 2^{501} gives

$$\boxed{2^{-501} \leqslant w < 2^{-500}.}$$

This does not determine w, but it gives us a clue: the unknown number is closely related to the upper limit 500 of the sum.

But is there anything mathematically special about 500? The summands follow the same pattern, regardless of where we stop. This suggests that 500 may not be essential to the problem.

Instead of trying to handle all 501 summands directly, let us replace 500 by a nonnegative integer m. Since the unknown number may depend on m, we denote it by w_m.

We are thus led to a more general problem: for each $m \in \mathbb{Z}_{\geqslant 0}$, find a real number w_m such that

$$\sum_{k=0}^{m} \left\lfloor \frac{x+2^k}{2^{k+1}} \right\rfloor = \lfloor x \rfloor - \lfloor w_m x \rfloor \qquad \text{for every } x \in \mathbb{R}. \tag{2}$$

At first sight, replacing 500 by an arbitrary m may seem to make the problem more difficult. However, this generalization allows us to investigate the simplest cases, starting with $m = 0$ and then $m = 1$.

Since the identity is required to hold for *every* real x, we can choose special values of x that simplify the floor expressions. These choices may help us discover a candidate for w_m, which we must then verify for all real x.

2 EXPLORING THE GENERALIZED PROBLEM

2.1 The case $m = 0$

Equation (2) becomes

$$\left\lfloor \frac{x+1}{2} \right\rfloor = \lfloor x \rfloor - \lfloor w_0 x \rfloor . \tag{3}$$

Step 1: Find a candidate.

Take $x = 2n-1$ with $n \in \mathbb{Z}$. For the argument below, we choose $n \geqslant 1$, so that $2n-1 > 0$. Then

$$\left\lfloor \frac{(2n-1)+1}{2} \right\rfloor = n, \qquad \lfloor 2n-1 \rfloor = 2n-1.$$

If (3) holds for all x, it follows that

$$n = 2n - 1 - \lfloor w_0(2n-1) \rfloor ,$$

and hence

$$\lfloor w_0(2n-1) \rfloor = n-1.$$

By the definition of the floor function,

$$n-1 \leqslant w_0(2n-1) < n.$$

Our choice $n \geqslant 1$ ensures that $2n-1 > 0$. Dividing by this positive number gives

$$\frac{n-1}{2n-1} \leqslant w_0 < \frac{n}{2n-1}.$$

Since the assumed identity holds for every real x, these inequalities hold for every positive integer n. Both bounds tend to $\frac{1}{2}$ as $n \to \infty$. Therefore,

$$\boxed{w_0 = \frac{1}{2}.}$$

This is a *necessary condition*. So far, we have not proved that this value works for every real x.

Step 2: Verify the candidate.

We recall the following special case of *Hermite's floor identity*:

$$\lfloor y \rfloor + \left\lfloor y + \frac{1}{2} \right\rfloor = \lfloor 2y \rfloor, \qquad y \in \mathbb{R}. \tag{4}$$

A proof is given in the appendix.

Taking $y = x/2$, we obtain

$$\left\lfloor \frac{x}{2} \right\rfloor + \left\lfloor \frac{x+1}{2} \right\rfloor = \lfloor x \rfloor .$$

Hence (3) holds for every real x with $w_0 = \frac{1}{2}$.

2.2 The case $m = 1$

Now the desired identity becomes

$$\left\lfloor \frac{x+1}{2} \right\rfloor + \left\lfloor \frac{x+2}{4} \right\rfloor = \lfloor x \rfloor - \lfloor w_1 x \rfloor . \tag{5}$$

We follow the same two-step strategy.

Step 1: Find a candidate.

As in the first case, take $x = 2n - 1$ with $n \in \mathbb{Z}$.

To simplify the second floor term as well, choose $n = 2s$ with $s \in \mathbb{Z}_{>0}$. Thus our test values are

$$x = 4s - 1 > 0.$$

Then

$$\left\lfloor \frac{x+1}{2} \right\rfloor = 2s, \qquad \left\lfloor \frac{x+2}{4} \right\rfloor = \left\lfloor s + \frac{1}{4} \right\rfloor = s.$$

Thus (5) implies

$$3s = 4s - 1 - \lfloor (4s-1) w_1 \rfloor ,$$

or equivalently,

$$\lfloor (4s-1) w_1 \rfloor = s - 1.$$

By the definition of the floor function,

$$s - 1 \leqslant (4s-1) w_1 < s.$$

Our choice $s \geqslant 1$ ensures $4s - 1 > 0$. Dividing by this positive number gives

$$\frac{s-1}{4s-1} \leqslant w_1 < \frac{s}{4s-1}.$$

Since the assumed identity holds for every real x, these inequalities hold for every $s \in \mathbb{Z}_{>0}$.

Letting $s \to \infty$, we obtain the necessary value

$$\boxed{w_1 = \frac{1}{4}.}$$

Step 2: Verify the candidate.

Apply (4) twice. With $y = x/4$,

$$\left\lfloor \frac{x}{4} \right\rfloor + \left\lfloor \frac{x+2}{4} \right\rfloor = \left\lfloor \frac{x}{2} \right\rfloor. \tag{6}$$

With $y = x/2$,

$$\left\lfloor \frac{x}{2} \right\rfloor + \left\lfloor \frac{x+1}{2} \right\rfloor = \lfloor x \rfloor. \tag{7}$$

Adding these equalities and cancelling the common term $\lfloor x/2 \rfloor$ gives

$$\left\lfloor \frac{x}{4} \right\rfloor + \left\lfloor \frac{x+2}{4} \right\rfloor + \left\lfloor \frac{x+1}{2} \right\rfloor = \lfloor x \rfloor.$$

This proves (5) for every $x \in \mathbb{R}$ with $w_1 = \frac{1}{4}$.

What have we discovered?

The first two cases give

$$w_0 = \frac{1}{2}, \qquad w_1 = \frac{1}{4},$$

suggesting that

$$w_m = \frac{1}{2^{m+1}}$$

might work in general.

More importantly, the proof for $m = 1$ reveals a structure: applying Hermite's identity at successive scales allows intermediate terms to cancel. This suggests a way to prove the conjecture for arbitrary m.

2.3 The general case

We now prove that the proposed formula works for every nonnegative integer m.

Theorem 2.1. *For every $m \in \mathbb{Z}_{\geqslant 0}$ and every $x \in \mathbb{R}$,*

$$\boxed{\sum_{k=0}^{m} \left\lfloor \frac{x + 2^k}{2^{k+1}} \right\rfloor + \left\lfloor \frac{x}{2^{m+1}} \right\rfloor = \lfloor x \rfloor.} \tag{8}$$

In particular, $w_m = 2^{-(m+1)}$ satisfies (2).

Proof. The case $m = 0$ has already been proved. We may therefore assume $m \geqslant 1$.

The key is again Hermite's identity

$$\lfloor y \rfloor + \left\lfloor y + \frac{1}{2} \right\rfloor = \lfloor 2y \rfloor.$$

Use it successively with

$$y = \frac{x}{2^{m+1}},\ \frac{x}{2^m},\ \dots,\ \frac{x}{4},\ \frac{x}{2}.$$

This gives

$$\begin{aligned}
\left\lfloor \frac{x}{2^{m+1}} \right\rfloor + \left\lfloor \frac{x+2^m}{2^{m+1}} \right\rfloor &= \left\lfloor \frac{x}{2^m} \right\rfloor, \\
\left\lfloor \frac{x}{2^m} \right\rfloor + \left\lfloor \frac{x+2^{m-1}}{2^m} \right\rfloor &= \left\lfloor \frac{x}{2^{m-1}} \right\rfloor, \\
&\vdots \\
\left\lfloor \frac{x}{4} \right\rfloor + \left\lfloor \frac{x+2}{4} \right\rfloor &= \left\lfloor \frac{x}{2} \right\rfloor, \\
\left\lfloor \frac{x}{2} \right\rfloor + \left\lfloor \frac{x+1}{2} \right\rfloor &= \lfloor x \rfloor .
\end{aligned}$$

Add all these equations. Every intermediate term $\left\lfloor x/2^j \right\rfloor$, $1 \leqslant j \leqslant m$, appears once on each side and cancels. What remains is

$$\left\lfloor \frac{x}{2^{m+1}} \right\rfloor + \left\lfloor \frac{x+2^m}{2^{m+1}} \right\rfloor + \left\lfloor \frac{x+2^{m-1}}{2^m} \right\rfloor + \cdots + \left\lfloor \frac{x+1}{2} \right\rfloor = \lfloor x \rfloor ,$$

which is exactly (8). □

Returning to the original problem, we take $m = 500$. Thus the required identity holds for every $x \in \mathbb{R}$ with

$$\boxed{w = \frac{1}{2^{501}}.}$$

3 REFLECTION: FROM A SPECIFIC PROBLEM TO A GENERAL PROOF

Our exploration began by testing a special value of x in the original problem. This gave us an interval for the unknown number w and suggested that its value might depend on the upper limit of the sum.

We then questioned whether the number 500 was essential. By replacing it with a nonnegative integer m, we turned a single problem into a family of related problems.

At first, this generalization might seem to make the problem more difficult. However, it allowed us to investigate the simplest cases, $m = 0$ and $m = 1$, which were not part of the original problem. Thus, generalizing the problem gave us a new way to explore it.

We then made another kind of simplification. For each fixed m, we chose special values of x to determine a candidate for w_m. These two choices served different purposes: the small values of m helped us understand the structure of the problem, while the special values of x helped us discover the unknown coefficients.

From these explorations, we conjectured that

$$w_m = \frac{1}{2^{m+1}}.$$

But discovering a pattern is not the same as proving it. The special cases suggested what might be true, while Hermite's floor identity allowed us to

prove that the formula works for every nonnegative integer m and every real number x.

This example illustrates how mathematical problem-solving can move between the specific and the general. Instead of attacking the original problem directly, we first explored a special case, generalized the problem, investigated simpler cases, formulated a conjecture, and finally established a general proof.

Sometimes, making a problem more general is precisely what makes it easier to understand.

A A PROOF OF HERMITE'S FLOOR IDENTITY

We prove that, for every $y \in \mathbb{R}$,

$$\boxed{\lfloor y \rfloor + \left\lfloor y + \frac{1}{2} \right\rfloor = \lfloor 2y \rfloor .}$$

Proof. Let $n = \lfloor y \rfloor$. Then we can write

$$y = n + r, \qquad 0 \leqslant r < 1.$$

Hence

$$\lfloor y \rfloor = n$$

and

$$\left\lfloor y + \frac{1}{2} \right\rfloor = n + \left\lfloor r + \frac{1}{2} \right\rfloor .$$

On the other hand,

$$\lfloor 2y \rfloor = 2n + \lfloor 2r \rfloor .$$

Since $0 \leqslant r < 1$, we distinguish two cases:

	$\left\lfloor r + \frac{1}{2} \right\rfloor$	$\lfloor 2r \rfloor$
$0 \leqslant r < \frac{1}{2}$	0	0
$\frac{1}{2} \leqslant r < 1$	1	1

Thus,

$$\left\lfloor r + \frac{1}{2} \right\rfloor = \lfloor 2r \rfloor ,$$

which gives

$$\lfloor y \rfloor + \left\lfloor y + \frac{1}{2} \right\rfloor = \lfloor 2y \rfloor .$$

□